

World's first rollable touchscreen tablet, inspired by ancient scrolls



A research team from Queen's University has taken a page from history, rolled it up and created MagicScroll—a rollable touchscreen tablet designed to capture the seamless flexible screen real estate of ancient scrolls in a modern-day device. The research was led by bendable-screen pioneer Dr Roel Vertegaal, professor of human-computer interaction and director of Queen's University Human Media Lab.

The device comprises a 19.05cm (7.5-inch), 2K resolution flexible display that can be rolled or unrolled around a central, 3D-printed cylindrical body containing the device's computerised inner workings. Two rotary wheels at either end of the cylinder allow the user to scroll through information on the touchscreen. To examine deeply, the display can be unrolled and function as a tablet display. Its lightweight and cylindrical body makes it easy to hold with one hand. When rolled up, it fits inside a pocket and can be used as a phone, dictation or pointing device.

"We were inspired by the design of ancient scrolls because their form allows for a more natural, uninterrupted experience of long visual timelines," says Dr Vertegaal.

Another source of inspiration was the old rolodex filing systems that were used to store and browse contact cards. MagicScroll's scroll wheel allows for infinite scroll action for quick browsing through long lists.

This new technology is set to push the boundaries of flexible device technology into brand new territory

Scientists increase Internet speed up to one-and-a-half times

Scientists from Samara University and University of Missouri (Columbia, Missouri) have developed an algorithm that provides fast and reliable access to powerful data processing centres for solving high-tech tasks. The algorithm, based on a special routing method, is relevant for solving problems that require high-precision calculations both in the field of fundamental science and for implementation of applied projects.

This is how it works. The user puts forward four basic requirements—certain signal bandwidth, speed of data transmission in Kbps, cloud storage and price of service. The algorithm gives solutions for specified criteria, while offering the best options for any network connection protocol, including fibre-optic networks or Wi-Fi.



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